MIL-V-19772A(SHIPS)
3 September 1968
SUPERSEDING
MIL-V-19772(SHIPS)
13 February 1957
(See 6.5)

MILITARY SPECIFICATION

VALVES, TEMPERATURE REGULATING,

AUTOMATIC, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope. - This specification covers automatic temperature regulating valves for steam, water and other fluid service where accurate temperature control of fluid is required. These valves are of the self-operated type requiring no external power supply such as oil, water, steam, electricity, or compressed gas for their operation.

1.2 Classification. - Automatic temperature regulating valves shall be of the following types, classes, and series as specified (see 6.2):

Type I - Direct operated 3-way valves-series 100.
Type II - Direct operated 2-way valves.
  Class 1 - Direct acting (heating service) - series 100, 150.
  Class 2 - Reverse acting (cooling service)- series 100, 150,250.
Type III - Pilot operated 2-way valves.
  Class 1 - Direct acting (heating service) - series 100, 150,400, 600.
  Class 2 - Reverse acting (cooling service) - series 100, 150,400, 600.
Type IV - Pilot operated, 2-way, combined temperature and pressure (heating service) series 100, 150, 400, 600 ---4:1 ratio and 1:1 ratio.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-D-1000/2	- Drawings, Engineering and Associated Lists.
MIL-F-1183	- Fittings, Tube, Cast Bronze, Silver Brazing.
MIL-R-2765	- Rubber Sheet, Strip, Extruded, and Molded Shapes, Synthetic, Oil Resistant.
MIL-M-15071	- Manuals, Equipment and Systems.
MIL-S-15083	- Steel Castings.
MIL-B-16541	- Bronze, Valve, Castings.
MIL-R-17131	- Rod, Welding, Surfacing.
MIL-V-17848	- Valve, Pressure Regulating, Steam Service.
MIL-F-20042	- Flanges, Pipe, Bronze, Silver Brazing.

STANDARDS

MILITARY

MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment.
MIL-STD-798	- Nondestructive Testing, Welding, Quality Control, Material Control and Identification and H.I.-Shock Test Requirements for Piping System Components for Naval Shipboard Use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 4820

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UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)
B1.12-1963 - Class 5 Interference-Fit Thread.
B16.5-1961 - Steel Pipe Flanges and Flange Fittings.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A216-1966 - Carbon-Steel Castings Suitable For Fusion Welding for High Temperature Service.
B61-1963 - Steam or Valve Bronze Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups, and using Federal agencies).

3. REQUIREMENTS

3.1 Qualification. - Automatic temperature regulating valves furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Definitions. - The following definitions are applicable to this specification.

3.2.1 Internal thermal element. - An element which is entirely enclosed within the valve body and which measures the temperature of the fluid passing through the valve.

3.2.2 External thermal element. - An element attached to the valve body by a suitable superstructure and to an externally located thermal bulb via capillary tubing.

3.2.3 Direct acting. - A 2-way valve is direct acting when an increase in the temperature being measured causes the valve plug to move toward the seated position. Generally used in heating service.

3.2.4 Reverse acting. - A 2-way valve is reverse acting when an increase in the temperature being measured causes the valve plug to move away from the seated position. Generally used in cooling service.

3.2.5 Fail safe. - A temperature regulating valve which incorporates a special thermal system which, in the event of leakage from the thermal element, positions the valve plug at the safe (as defined below) end of its travel:

3-way valve (type I) - Valve plug position resulting from failure of the thermal element will divert at least 95 percent of the flow to the cooler.
2-way valve (types II or III) - Direct acting valves used for heating to fail closed and shut off steam flow. Reverse acting valves used for cooling service to fail open to pass full cooling fluid flow.

3.2.6 Dead band. - Dead band is a measure of the inability of the valve to assume an identical position for the same temperature on its upward and downward travel.

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3.2.7 Control band. - The temperature change required to move the valve plug through its full travel from the shut-off to the fully open position or vice versa. Influences such as internal friction and pressure drop forces must be taken into account when determining the control band for a particular application. "Accuracy of regulation" is the amount, under a steady state condition, by which the controlled temperature deviates from the set temperature. The "accuracy of regulation" will be finer than the control band of that valve, in cases where the valve plug does not need to move through its full travel to meet the range of load demands of a particular application.

3.2.8 Set temperature. - The nominal setting of the valve. Usually taken to be the mid-point of the range within which the temperature is required to be controlled or the temperature at the mid-point of the valve plug travel.

3.2.9 Adjustment limits. - The range of set temperatures over which the valve can be adjusted while meeting the performance requirements.

3.2.10 Internal bellows. - A bellows which is normally in contact with the process fluid.

3.2.11 External bellows. - A bellows which is normally external to the body and is not in contact with the process fluid.

3.3 Materials of construction. - Materials shall be as specified in table I and 3.3.1 through 3.3.2. All materials shall be selected to prevent galling, seizing, or excessive wear between parts. Clearance shall be such as to prevent interference due to thermal expansion. This list of materials is not intended to be restrictive, providing it is demonstrated that proposed alternate materials are satisfactory. The Naval Ship Engineering Center (NAVSEC) (Auxiliary Equipment Branch) approval is required for the use of any alternate materials.

Table I - List of materials.

Name of parts	Nonferrous (series 100 and 250)	Ferrous (series 150, 400, and 600)
Body, bonnet, and pilot body (where separate).	Bronze MIL-B-16541 ASTM-B61	Carbon steel MIL-S-15083, class B ASTM-A216, grade WCB
Springs (external)	Carbon steel - cadmium plated	
Springs (internal)	Nickel-copper alloy, Nickel-chrome alloy, Inconel, Corrosion-resisting steel (300 series)	
Bellows (external)	Brass	
Bellows (internal)	Bronze, Nickel-copper alloy	
Diaphragm (metallic)	Phosphor bronze, Nickel-copper alloy, Corrosion-resisting steel (300 series)	
Diaphragm (nonmetallic)	Gasket-oil resisting MIL-R-2765	

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3.3.1 Internal trim. - The internal trim (valve stem, seat ring and plug) for ferrous valves (series 150, 400, and 600) shall be corrosion-resisting steel. Unless otherwise specified (see 6.2), the seating and flow controlling surfaces on the seat ring and disc (main valve only) shall be hardfaced (see 3.4.11). Materials used for the internal trim of nonferrous valves (series 100 and 250) shall be as follows: For valves used in fresh water service, materials shall be bronze, corrosion-resisting steel, and nickel-copper alloy. For valves used in steam service, valve discs shall be of 400 series corrosion-resisting steel (hardened to 500 Brinell minimum) and seats shall be hardfaced. For valves used in sea water service, material shall be nickel-copper alloy. The specification for nickel-copper alloy is not intended to limit the choice to basic monel alloy, but rather is intended to permit the use of the most suitable grade of nickel-copper alloy for each particular application (for example, S-monel and K-monel would be considered to be nickel-copper alloy). The proper grades of the materials specified above, for use in both ferrous and nonferrous valves, shall be chosen to have the necessary strength, hardness, and nongalling properties to prevent distortion, galling, or excessive wear between operating parts.

3.3.2 External thermal bulb. - The material for the external thermal bulb shall be copper for fresh water, copper or carbon steel for lube oil service, carbon steel or corrosion-resisting steel for fuel oil service, and nickel-copper alloy for sea water service.

3.4 Design and construction. -

3.4.1 Pressure-temperature ratings. - The pressure-temperature ratings shall be as specified in table II.

Table II - Pressure-temperature ratings.

Series	Body Material	Maximum allowable pressure (psig)	Maximum allowable temperature (°F.)
100	Bronze	100	425
250	Bronze	250	150
150	Steel	150	500
400	Steel	400	775
600	Steel	600	775

3.4.2 End connections. - End connections shall be integral with the body and in accordance with table III. Union end valves shall be furnished complete with union nuts and tailpieces.

Table III - End connections.

Series	Flanged end 1/	Union end
	Reference documents	
100	MIL-F-20042	MIL-F-1183
250	MIL-F-20042	MIL-F-1183
150	USAS B16.5	----------
400	USAS B16.5	----------
600	USAS B16.5	----------

1/ Flange bolting shall straddle the vertical center line. When an odd number of bolts is used, the vertical center line at the top of the flange shall be straddled.

3.4.2.1 Type II, series 100 and 250, valves in sizes up to and including 2 inches shall be union end. Type I valves in sizes up to and including 2 inches may be union end or flanged (see 6.2). All other valves shall be flanged. The 1/4 and 3/8 inch type II valves shall be furnished with 1/2 inch end connections.

3.4.3 Accessibility. - With the exception of type I valves, all valves shall be accessible for adjustment, repair, or replacement of all internal trim, including the thermal element, without requiring removal of the valve from the line.

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3.4.4 Capillary tubing. - The capillary tubing shall be covered throughout the entire length with a flexible bronze, brass, or corrosion-resisting steel armor so constructed that the flexibility of the tubing will not be excessively reduced. At the point where the tubing joins the bulb and the valve, the tubing shall be reinforced with an additional layer of armor for a minimum distance of 5 inches, to prevent kinking or mechanical failure at this point. The armor shall be smoothly finished and free from all mechanical defects. Unless otherwise specified in the contract or order, the length of the capillary tubing between the bulb and the valve body shall be 10 feet.

3.4.5 Thermal media. - The media employed in the thermal element shall have no detrimental effect on the material with which it comes in contact, and shall have no detrimental effect to personnel who may be exposed to it upon leakage or failure of the element.

3.4.6 Bellows. - Bellows, where employed, shall be of a seamless type. Connections shall be hermetically sealed by interlocking and soldering, welding or brazing. Bellows may be either single or double ply (laminated construction).

3.4.7 Bulb. - The bulb shall be either of one piece construction or shall be joined together by welding or silver brazing so that the joint is comparable to one piece construction. Bulb connection shall be flanged. Flange rating to be as specified in 6.2. Bulb length shall not exceed 22 inches for valves in sizes up to and inclusive of 1-1/2 iron pipe size (i.p.s.). For valves in sizes 2 inches and above, bulb length shall be kept to a minimum.

3.4.8 Thermal element. - All type II, III and IV valves shall be furnished with an adjustable external thermal element (see 3.5.1). Type I valves shall be furnished with either an adjustable external thermal element or an adjustable or fixed setting internal thermal element (see 6.2). Thermal elements shall be interchangeable between temperature regulating valves of the same type, class, series, size, and manufacture. Thermal elements for type IV valves shall be completely interchangeable between all sizes of valves of a manufacturer for any given temperature range.

3.4.9 Overange protection. - Valves furnished with thermal elements whose service range does not extend over 100°F. shall be capable of withstanding a temperature at least 50°F. above the upper range limit without detrimental effect to any part of the valve. Valves furnished with thermal elements whose service range exceeds 100°F. shall be capable of withstanding a temperature at least 20°F. above the upper range limit without detrimental effect to any part of the valve. Type I valves furnished with internal thermal elements shall be capable of withstanding a temperature of 215°F. without detrimental effect to any part of the valve.

3.4.10 Safety features. -

3.4.10.1 Type I valves. - Type I valves shall incorporate at least one of the following safety features:

(a) Loss of the thermal charge will result in a valve plug position that directs at least 95 percent of the flow to the cooler.
(b) Provision to manually regulate flow in the event of loss of the thermal charge.
(c) Provision of at least two independent and interchangeable thermal elements each capable of controlling the regulating valve at all times.

3.4.10.2 Type II and III valves. - Where danger or injury to personnel or damage to equipment may occur in the event of an oversupply of heating steam or loss of cooling fluid, type II or III valves shall be of fail safe construction as defined in 3.2.5.

3.4.10.3 Type IV valves. - In the event of loss of the thermal charge, the thermal feedback signal shall be removed from the regulating valve and it shall function as a pressure reducing valve. The reduced pressure setting shall be manually adjustable with the valve under pressure.

3.4.11 Seats.- Valve seats shall be replaceable except for type III and IV valves where seats may be separate or integral with the body. Valves without separate seat rings shall have sufficient metal thickness at the valve seat to permit the installation of a separate seat ring if required as a service repair. Threaded in seat rings shall be seal welded circumferentially on valves to be used on steam service above 225 psig. Where seating surfaces are hardfaced, the inlay shall be in accordance with MIL-R-17131 and shall have a minimum thickness of 3/32 inch. All type IV valves shall have hardfaced seats.

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3.4.12 Plug guiding. - The stem/plug assembly guiding shall prevent binding or seizing due to lateral thrust, thermal expansion, or body distortion caused by internal pressure or the forces and moments imposed by the connecting piping, or any combination of these, and insure proper seating. This requirement shall be maintained with interchangeable parts and under all tolerance stack-up conditions.

3.4.13 Bolting. - Flange bolting, for other than end connections, shall be one of the following:

(a) Through bolts fitted with a nut on each end. Threads on bolts and nuts shall be class 2 fit.
(b) Tap-end studs with interference fit at the tap end and a class 2 fit at the nut end. Interference fit shall be in accordance with USAS B1.12.

Bearing surfaces of nuts and bolts shall be finish machined. Their respective mating surfaces on the valve shall also be finish machined if not cast smooth and true.

3.4.14 Threads. - All threads shall conform to Handbook H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used.

3.4.15 Springs. - Springs shall be designed so that they will not be fully compressed during any normal operation or adjustment of the valve. The working stress shall be such that relaxation shall not exceed 5 percent over a 1000 hour period at the operating temperature. Spring ends shall be squared and ground.

3.4.16 Interchangeability. - All parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation and performance without requiring selection, fitting, or machining of any kind.

3.4.17 Set point adjustment. - Means shall be provided for adjusting the temperature set point over the range specified in table IV with the valve under pressure. Means shall also be provided on type IV valves for adjusting the outlet pressure through a range not less than 75 percent to 125 percent of the mid-range set pressure with the installed spring, without requiring replacement of any internal parts. A locknut or other suitable device shall be provided to prevent accidental change of any adjustments.

3.4.18 Diaphragms. - Where type III valves are of the diaphragm actuated type, and where the diaphragm is nonmetallic, the valve shall be designed to limit the temperature of the diaphragm to 180°F. maximum at all times, with an ambient temperature of 110°F.

3.4.19 Type IV valve requirements. - Type IV valves shall be basically internal pilot operated pressure reducing valves in accordance with class A of MIL-V-17848 with the addition of the provision which automatically modifies the reduced pressure setting as a function of the downstream temperature by means of a temperature signal feedback circuit into the pilot section.

3.5 Performance requirements. -

3.5.1 Temperature range and accuracy of regulation. - Valves shall be capable of maintaining the controlled temperature within the limits specified in table IV. Minimum ranges for adjustable thermostatic elements and the service temperature range for which elements must be supplied shall be as specified in table IV.

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Table IV - Temperature range and accuracy of regulation.

Type	Class	Adjustment limits °F. (minimum)	Service temperature range °F.	Control band °F.
I	-----	Fixed	100-180	±5
I	-----	30	100-180	±7.5
II	1	30	60-200	±7.5
II	2	30	60-160	±7.5
III	1 and 2	40	60-260	±3
IV	-----	40	60-260	1/ ±7.5

1/ Control band to be determined by set pressure ratio test specified in 4.2.6.1.

3.5.2 Ambient temperature effect. - Change in ambient temperature around the capillary tubing and valve from 80°F. to 110°F. shall not influence the controlled fluid temperature by more than 4°F. The valve shall continue to function when the ambient temperature at the bellows is 140°F.

3.5.3 Dead band (type II only). - Measured in percentage of valve stroke, dead band shall not exceed 5 percent.

3.5.4 Seat tightness. - With the exception of types I and IV, valves shall provide lock-up at the control band limit. Type I valves may have a maximum of 5 percent leakage. Type IV valves shall provide positive lock-up within the dead-end build-up limits specified in 4.2.6.3.

3.5.5 Stability. - Valves shall operate over the entire flow range without hunting, chattering, or creating excessive noise and vibration, at any setting within the adjustment range.

3.5.6 Capacity. - Minimum required valve capacities shall be as specified in 3.5.6.1 through 3.5.6.6.

3.5.6.1 Type I valve capacities. - Minimum water capacities for type I valves with a 5 psi drop across the valve and with all flow through the cooler port shall be in accordance with table V.

Table V - Capacity for type I valves.

Nominal pipe size (i.p.s.)	Flow (g.p.m.)
Inches	
1/2	6
3/4	10
1	15
1-1/4	20
1-1/2	25
2	86
2-1/2	125
3	185
4	350
5	450
6	600

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3.5.6.2 Type II, class 1 valve capacities. - Minimum capacities for type II, class 1 valves based on a 35 psig steam inlet pressure and an outlet pressure of 20 psig or less, and with the valve set in the mid-point of its adjustable range and with the element sensing a temperature of 15°F. below the temperature at which the valve is set to close, shall be in accordance with table VI.

Table VI - Capacity for type II, class 1 valves.

Nominal pipe size (i.p.s.)	Capacity (pounds steam per hour)
Inches	
1/4	40
3/8	80
1/2	100
3/4	300
1	500
1-1/4	1000
1-1/2	1300
2	2000
2-1/2	3000
3	4500
4	7500

3.5.6.3 Type II, class 2 valve capacities. - Minimum water capacities for type II, class 2 valves with a 5 psi drop across the valve and with the element sensing a temperature of 15°F. above the temperature at which the valve is set to close, shall be in accordance with table VII.

Table VII - Capacity for type II, class 2 valves.

Nominal pipe size (i.p.s.)	Capacity (g.p.m.)
Inches	
1/4	1.4
3/8	2.1
1/2	7
3/4	10
1	13
1-1/4	20
1-1/2	30
2	55
2-1/2	80
3	125
4	200

3.5.6.4 Type III, class 1 valve capacities. - Minimum capacities for type III, class 1 valves based on a 35 psig steam inlet pressure and an outlet pressure of 20 psi or less, and with the element sensing a temperature of 6°F. below the temperature at which the valve is set to close, shall be in accordance with table VIII.

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Table VIII - Capacity for type III, class 1 valves,

Nominal pipe size (i.p.s.)	Capacity (pounds steam per hour)
Inches	
1/2	150
3/4	300
1	500
1-1/4	1000
1-1/2	1300
2	2000
2-1/2	3000
3	4500
4	7500

3.5.6.5 Type III, class 2 valve capacities. - Minimum water capacities for type III, class 2 valves with a 5 psi drop across the valve and with the element sensing a temperature of 6°F. above the temperature at which the valve is set to close, shall be in accordance with table IX.

Table IX - Capacity for type III, class 2 valves.

Nominal pipe size (i.p.s.)	Capacity (g.p.m.)
Inches	
1/2	7
3/4	10
1	20
1-1/4	30
1-1/2	40
2	60
2-1/2	110
3	150
4	230

3.5.6.6 Type IV valve capacities. - Minimum capacities for type IV valves based on a 100 psig steam inlet pressure and an outlet pressure of 55 psig, or less, and with the element sensing a temperature of 6°F. below the temperature at which the valve is set to close, shall be in accordance with table X.

Table X - Capacity for type IV valves.

Nominal pipe size	Saturated steam capacity pounds per hour
Inches	
1/2	275
3/4	510
1	850
1-1/4	1,520
1-1/2	2,100
2	3,500
2-1/2	5,000
3	7,800
3-1/2	10,500
4	13,500

3.6 Mechanical shock and vibration. - The valves shall be designed to withstand shock and vibration. When specified (see 6.2), shock tests shall be in accordance with grade A, class I of MIL-S-901 and vibration tests shall be in accordance with type I of MIL-STD-167.

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3.7 Marking. -

3.7.1 Body markings. - The manufacturer's name or trademark shall be cast or forged integral with the valve body. For type II, III and IV valves, a flow arrow shall be permanently marked on the body. For type I valves, the inlet connection shall be marked "inlet", the bypass connection shall be marked "bypass", and the cooler connection shall be marked "cooler".

3.7.2 Identification plates. - Each valve shall have an identification plate permanently attached to an exposed position on the valve that will not be covered by insulation. The identification plate shall be made of a corrosion-resistant material and shall contain the following information or a space therefor:

(a) Manufacturer's name or trademark.
(b) MIL-V-19772 and type, class, series, and if of fail safe construction.
(c) Size and capacity.
(d) Temperature range of element.
(e) Body and trim material composition.
(f) Manufacturer's drawing number.
(g) Manufacturer's model and part number or identification.
(h) Applicable manual number.
(i) Space for 9 digit CID number.

3.8 Drawings. -

3.8.1 Preliminary drawings. - Preliminary drawings which are sufficient to permit evaluation of the design and approval of materials, shall be submitted with bids to the procuring activity. These drawings shall show the following:

(a) Accurately scaled sectional assembly which clearly depicts the design and construction of the valve.
(b) Bill of materials listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.
(c) Dimensions-overall, accessibility space, including disassembly clearances, and all other dimensions pertinent to installation.
(d) Estimated weight and any limitations on installation of the valve.
(e) Reference to any previous shock and vibration approval for the valve and test report numbers.
(f) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.

3.8.2 Final drawings. - Final drawings and certification data sheets shall be submitted to the procuring activity for approval within 60 days after date of contract. These drawings shall be in accordance with types II and III of MIL-D-1000/2 except for extent of detail. Only the information required in 3.8.1 need be furnished for type II drawings. The following data, in addition to that required in MIL-D-1000/2, shall be furnished for the type III drawings:

(a) Ship identification.
(b) Applicable assembly drawing number(s).
(c) Applicable manual number.
(d) CID (APL) number.
(e) Application description.
(f) Valve description.
(g) Set point and adjustable range.

3.8.3 Limited rights legend. - When the Government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

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"Furnished under United States Government Contract No._____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of_____. except for: (a) emergency repair or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

3.9 Manuals. - Manuals shall be furnished in accordance with type I of MIL-M-15071. The quantity and distribution of manuals shall be as specified (see 6.2). The following, in addition to that required for type I of MIL-M-15071, shall be included as part of the manual contents:

(a) The approved engineering drawings for the valve (including certification data sheet). These drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, etc., and may eliminate extraneous lines and details to clarify the interaction of parts.
(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.
(c) Instructions to permit overhaul by shipyard or other repair facility. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, etc. Also the appropriate procedure (i.e. part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.
(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.
(e) Adjustment procedures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to precribed requirements.

✓ 4.2 Qualification tests[1/]. - Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.2.2 through 4.2.6.

4.2.1 Qualification test sample. - The following categories shall require separate qualification testing:

Type I
Type I (fail safe construction)
Type II, class 1
Type II, class 1 (fail safe construction)
Type II, class 2
Type II, class 2 (fail safe construction)
Type III, class 1
Type III, class 1 (fail safe construction)
Type III, class 2

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

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Type III, class 2 (fail safe construction)
Type IV, 4:1 ratio
Type IV, 1:1 ratio

4.2.1.1 Qualification of the size and series of valve tested will include qualification of all other valves in the tested category for all sizes and for all series up to and including the series of the test valve. The size of the test valve shall be as approved by NAVSEC.

4.2.1.2 Detailed engineering drawings of the test valve, and assembly drawings of all other sizes and series of that category valve for which qualification is desired, shall be submitted with the test valve.

4.2.2 Visual examination. - A visual examination shall be made to verify conformance to the requirements of this specification.

4.2.3 Tests for type I valves. -

4.2.3.1 Responsiveness. - The thermal element shall be inserted in fluid at the temperature at which all of the flow is directed through the by-pass circuit. Raise the temperature of the fluid acting on the thermal element instantaneously by 30°F. Under the new temperature condition, the valve shall reposition itself to flow not less than 95 percent of the inlet water to the cooler outlet within 30 seconds.

4.2.3.2 Flow test. - With the thermal element sensing a temperature equal to the upper limit of the control band (see 3.5.1) the water flow rate through the cooler port shall be determined with a 5 psi pressure drop across the valve. Minimum flow capacity shall be in accordance with 3.5.6.1.

4.2.4 Tests for type II valves. -

4.2.4.1 Responsiveness. -

4.2.4.1.1 Class 1 valves. - With steam at the design operating pressure and temperature and a flow in accordance with table VI through the valve, the temperature at the thermal element should be instantaneously raised 25°F. The valve should fully close within 40 seconds.

4.2.4.1.2 Class 2 valves. - With cooling water at the design operating pressure and temperature and a flow of maximum capacity through the valve in accordance with table VII, the temperature at the thermal element should be instantaneously dropped 20 degrees below the temperature at which the valve is fully open. The valve should fully close within 40 seconds.

4.2.4.2 Flow test. - The valve shall be tested when set at the lower, mid-point, and upper limits of its adjustable range. At each setting, the temperature which the element is sensing shall be varied from the upper to lower limits of the control band specified in 3.5.1 and the fluid flow (heating or cooling as applicable) shall vary from lock-up to the minimum required flow specified in 3.5.6.2 or 3.5.6.3 under the pressure conditions specified in 3.5.6.2 or 3.5.6.3.

4.2.5 Tests for type III valves. -

4.2.5.1 Responsiveness. -

4.2.5.1.1 Class 1 valves. - With steam at the design operating pressure and temperature and a flow of maximum capacity through the valve in accordance with table VIII, the temperature at the thermal element should be instantaneously raised 20°F. above the temperature at which the valve is fully open. The valve should fully close within 15 seconds.

4.2.5.1.2 Class 2 valves. - With cooling water at the design operating pressure and temperature and a flow of maximum capacity through the valve in accordance with table IX, the temperature at the thermal element should be instantaneously dropped 20 degrees below the temperature at which the valve is fully open. The valve should fully close within 15 seconds.

4.2.5.2 Flow test. - The valve shall be tested when set at the lower, mid-point, and upper limits of the adjustable range. At each setting, the temperature which the element is sensing shall be varied from the upper to lower limits of the control band specified in 3.5.1 and the fluid flow (heating or cooling as applicable) shall vary from lock-up to the minimum required flow specified in 3.5.6.4 or 3.5.6.5 under the pressure conditions specified in 3.5.6.4 or 3.5.6.5.

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4.2.6 Tests for type IV valves. -

4.2.6.1 Ratio test. - For valves with a 4:1 ratio, set the pressure limiting device to deliver 60 psig. Set the temperature adjustment lower limit at 150°F. (so that 150°F. just starts to drop the reduced pressure). Subject the thermal element to 165°F. water and check that the reduced pressure drops to 0 psig. For valves with a 1:1 ratio, conduct as above, except set the pressure limiting device to deliver 15 psig and check that the reduced pressure drops to 0 psig with an increase of 15°F.

4.2.6.2 Pressure limiting test. - Drop the temperature which the temperature element is sensing at least 15°F. below the lower limit of the temperature control band, and check that the reduced pressure does not exceed the limit setting.

4.2.6.3 Pressure regulation test. - The pressure regulation test shall be conducted with the thermal system maintained constant under two conditions. First at the lower control band limit (where the thermal feedback just begins to limit the reduced pressure) and second at approximately the mid-point of the control band (where the thermal feedback decreases the reduced pressure setting to approximately half of the limit setting). Under each condition, vary steam flow from lock-up to maximum as specified in 3.5.6.6 by varying the downstream demand. The reduced pressure shall be maintained within the limits of plus or minus 5 percent or plus or minus 2 psi, whichever is greater. Dead-end buildup shall not exceed 10 psi in a one hour period with a downtream volume not exceeding the volume represented by 100 diameters of downstream pipe.

4.3 Sampling for quality conformance inspection. -

4.3.1 Lot. - All valves of the same type, class, series, size, thermal element and adjustable range, offered for delivery at one time, shall be considered a lot for the purpose of sampling.

4.3.2 Sampling for visual and dimensional examination. - A random sample of valves shall be selected from each lot in accordance with table XI and shall be examined as specified in 4.4.1.1 and 4.4.1.2. Failure of any valve in the sample to pass the examination specified in 4.4.1.1 and 4.4.1.2 shall be cause for rejection of the lot.

Table XI - Sampling for visual and dimensional examination.

Lot size	Sample quantity
2 to 25	1
26 to 65	2
66 to 180	3
Over 180	4

4.3.3 Sampling for tests. - A random sample of valves shall be selected from each lot in accordance with table XII and shall be tested as specified in 4.4.2. If the number of rejected valves in any sample exceeds the acceptance number specified in table XII the lot represented by the sample shall be subject to rejection. If the sample size specified in table XII equals or exceeds the lot size, the lot shall undergo 100 percent inspection.

Table XII - Sampling for tests.

Lot size	Sample size	Allowable number of rejects
2 to 8	5	0
9 to 15	7	0
16 to 25	10	0
26 to 40	15	0
41 to 65	25	0
66 to 110	35	1
or over		

4.4 Quality conformance inspection. -

4.4.1 Examination. -

4.4.1.1 Visual examination. - A visual examination shall be made of the sample valves selected in accordance with 4.3.2 to verify conformance to the requirements of this specification.

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4.4.1.2 Dimensional examination. - A dimensional examination shall be made on the sample valves selected in accordance with 4.3.2 to verify conformance with the approved drawings.

4.4.2 Tests. -

4.4.2.1 Nondestructive tests. - Nondestructive tests (radiography, magnetic particle, and dye penetrant) shall be conducted in accordance with MIL-STD-798.

4.4.2.2 Pressure tests. - Each sample valve selected in accordance with 4.3.3 shall be pressure tested in accordance with table XIII. (It should be noted that control diaphragms in type III valves are not designed to withstand hydrostatic test pressures; therefore, prior to testing, the diaphragms should be removed or the pilot parts should be plugged to prevent diaphragm rupture.) Visible deformation or porosity or other signs of weakness under the hydrostatic test shall be cause for rejection.

Table XIII - Pressure tests hydrostatic p.s.i. - for strength and porosity.

	Series				
	100	150	250	400	600
With the valve open	P.s.i.g. 200	P.s.i.g. 425	P.s.i.g. 500	P.s.i.g. 1450	P.s.i.g. 2175

4.5 Inspection of preparation for delivery. - The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Sub-contracted material and parts. - The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

5.2 Domestic shipment and early equipment installation (see 5.4.1). -

5.2.1 Valves. -

5.2.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice. All ports shall be plugged or sealed with approved corrosion-resistant material to prevent entrance of moisture and dirt.

5.2.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.3 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.4.2 and 6.2).

5.4 Use of polystyrene (loose-fill) material.

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5.4.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.4.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. -

6.1.1 Type I. - Type I valves are intended for use in diesel engine cooling service and other water mixing services where proportioning of water to a cooler and a by-pass is required.

6.1.2 Type II. - Type II valves are intended for use with hot water heaters and other apparatus where steam or hot water is used to heat water and for condensate recirculation and general cooling services.

6.1.3 Type II, series 250. - Type II, series 250 valves are intended primarily for use in cooling service where sea water is the cooling fluid.

6.1.4 Type III. - Type III valves are intended for services where fine control over a narrow range is required and where direct operated valves are not suitable for the pressures, temperatures and response requirements.

6.1.5 Type IV. - Type IV valves are intended for use on hot water heaters, fuel oil heaters and for any other apparatus where steam is used as the heating medium. Type IV valves are intended for use where close control is required and on services like instantaneous heaters where direct operated valves and type III valves are not suited.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and series required (see 1.2).
(c) Whether a special fail safe construction is required for type I valves (see 3.2.5 and 3.4.10.1(a).
(d) Whether a special fail safe construction is required for type II or III valves (see 3.2.5).
(e) Whether hardfacing is required for internal trim of ferrous valves (see 3.3.1).
(f) End connections required (see 3.4.2.1).
(g) Bulb flange rating (see 3.4.7).
(h) Type of thermal element required for type I valves (see 3.4.8).
(i) Size required (and capacity, if other than as specified in 3.5.6).
(j) Temperature setting.
(k) Reduced pressure setting and ratio required for type IV valves.
(l) Maximum and minimum inlet pressure and temperature conditions.
(m) Brief description of application.
(n) When shock and vibration tests are required (see 3.6).
(o) Manuals (quantity and distribution)(see 3.9).
(p) Level of preservation, packaging, packing and marking required if other than as specified in 5.2 (see 5.3).
(q) When polystyrene (loose-fill) material is approved (see 5.4.1).

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6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 19772 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Interchangeability relationship. - The valves covered by this specification are not interchangeable with those previously covered. This specification changes the end connection requirements to bring them up to current Navy practice.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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